

Lovesh Kumar

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EDUCATION

University of South Florida – Tampa, FL

May 2026

B.S. Computer Science | GPA: 3.66/4.0

Relevant Courses: Operating Systems, Data Structures & Algorithms, Analysis of Algorithms, Software System Development, Advanced OOP, Automata Theory

EXPERIENCE

USF Water Institute – Database Application Intern

May 2025 – Aug 2025

- Designed and implemented a **RESTful API** in **.NET Core (Fast Endpoints)** to serve water quality data across **40+ parameters**, reducing average response time by **30%** through server-side pagination and query optimization.
- Used **Entity Framework Core** with **LINQ** to query a production **SQL database**, abstracting complex multi-parameter joins and enabling scalable data delivery to external clients.
- Refactored backend with **layered services and DTO mappings**, reducing maintenance overhead by **30%** and decoupling business logic from API contracts for easier feature expansion.
- Managed **CI/CD pipelines** using **Git** and **Azure DevOps**, automating build, test, and deployment workflows to improve release reliability.

PROJECTS

MemeCrafter – Meme Analysis System | Python, PyTorch, BLIP-2, LoRA, EasyOCR, Gradio

- Fine-tuned **BLIP-2 (2.7B parameter vision-language model)** on a custom meme dataset using **LoRA adapters**, enabling meme-specific captioning and sentiment classification.
- Built a **multi-modal analysis pipeline** combining **EasyOCR** text extraction with BLIP-2 visual understanding to detect sentiment (positive, negative, neutral, sarcastic) with context-aware accuracy.
- Deployed an interactive **Gradio web app** for real-time meme analysis, including a side-by-side base vs. fine-tuned model comparison view.

ASL Gesture Recognition System | Python, TensorFlow, MediaPipe, CNN, OpenCV

- Built a complete **CNN-based pipeline** to classify **36 ASL signs (A–Z and 0–9)** in real time, using **MediaPipe** for hand landmark detection and background segmentation.
- Trained multiple CNN architectures with **data augmentation** (rotation, zoom, shear), **batch normalization**, and dropout, achieving **88–93% test accuracy** across 36 classes.
- Deployed a **live webcam inference system** using **OpenCV** with prediction smoothing and confidence scoring, running at **15–30 FPS**.

Mental Health Chatbot | Python, TensorFlow, Flask, React, TypeScript, PostgreSQL, AWS

- Developed a **full-stack AI chatbot** with a **React/TypeScript** frontend and **Flask** backend, using a hybrid **TensorFlow NLP** model + keyword matching to classify 7 intents with **95%+ validation accuracy**.
- Trained a **sequential neural network** (Embedding → GlobalAveragePooling → Dense) with early stopping and learning rate scheduling, improving **F1 score from 0.50 to 0.60** through iterative data expansion.
- Secured user data with **AWS Cognito** authentication and built a **PostgreSQL-backed** session logging system to enable continuous model retraining from new interaction data.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, HTML/CSS, RISC-V

Frameworks & Libraries: .NET Core (Fast Endpoints), Flask, TensorFlow, PyTorch, OpenCV, MediaPipe, React.js, Node.js

Cloud & Tools: AWS (Lambda, S3, Cognito), Azure DevOps, Git, Linux, PostgreSQL, Entity Framework Core, CI/CD, Gradio

LEADERSHIP & ACTIVITIES

USF Bellini College of AI, Cybersecurity & Computing – CS Tutor

2025 – Present

- Tutor students in **Data Structures, Intro to Programming, Program Design**, and **Computer Architecture** through one-on-one and small group sessions.
- Support students with **code debugging, algorithm analysis**, and exam preparation, improving comprehension and academic performance.

Bhakti Yoga Society – USF – President

2023 – Present

- Lead a **50+ member** student organization, growing membership **20% YoY** through expanded programming and university department partnerships.
- Manage a **6-person officer team** overseeing weekly event logistics, budget allocation, and campus outreach.

Society of Asian Scientists & Engineers (SASE) – Conference Chair

2025 – Present

- Oversaw planning and execution of the **2025 SASE Regional Conference**, coordinating **5+ corporate partners** and university departments to deliver career panels, networking sessions, and technical workshops for **200+ STEM students** across multiple universities.
- Led a **10-member team** across logistics, marketing, and budgeting using structured project management, achieving a **95% attendee satisfaction rate**.